



# GRAPHIC PORTFOLIO



Jonathon Simpson

## Web App Design & Development

Each example shows one or more screenshots of actual, working, full-stack applications, along with descriptions of UI/UX and ancillary details about the tech stack. All designs and development by myself, from start to finish, front to back, inside and out, most of them comprising deep UX considerations for interaction with complex systems using intricate workflows for technical applications.

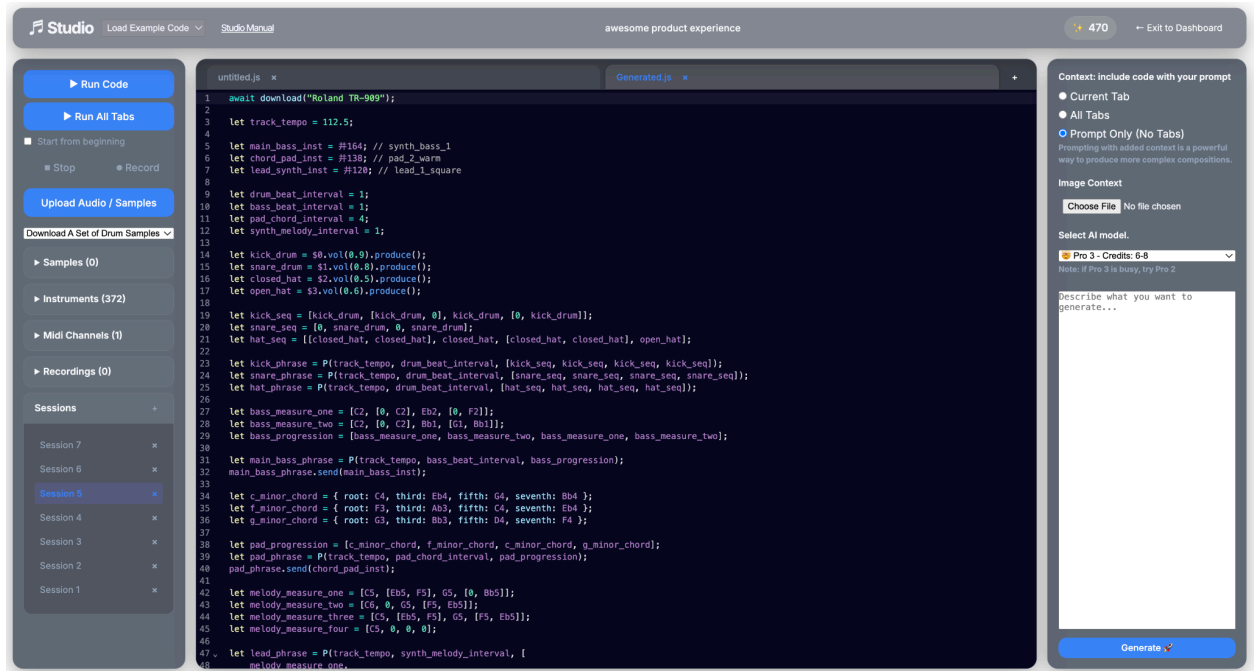
### Table of Contents

1. **Music Programming Studio** - My most recent work (2026)
  - a. Front Page, Registration and Login
  - b. Dashboard & Payments
  - c. Full Studio App
  - d. Quick Prompt App
  - e. Mobile Studio App
2. **Chess Foo** - My first full stack application (2013)
3. **G-tone Scientific Synthesizer** - Gaussian tuned sounds (2020)
4. **P2P Audio Broadcasting** - Open Source Project (2022)
5. **Figma Designs** (2021)
  - a. Mobile Website
  - b. Tablet Application

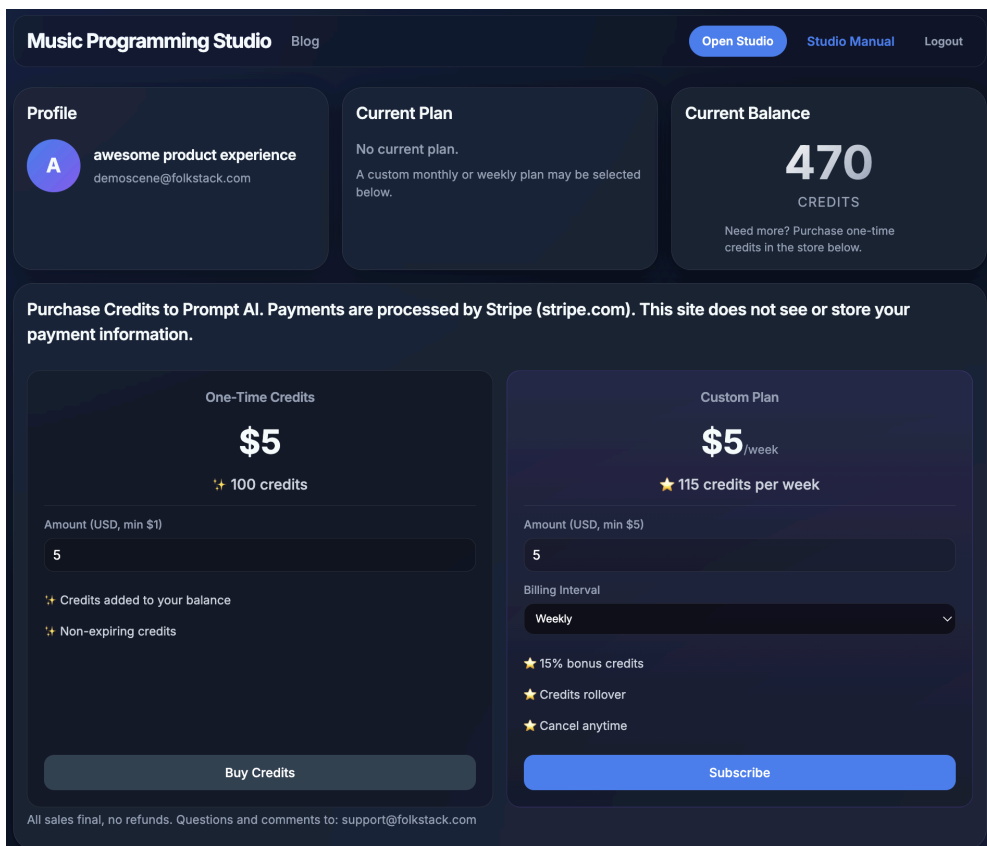
### 1. Music Programming Studio and Website

The Music Programming Studio is a total product. It has user registration, login, payments, a dashboard, and several interfaces. At the core of the studio app is the engine, which translates code into music, and provides many high level functions designed to make the programming of music akin to composing with bars and measures. The studio now includes Agent prompting to generate code for the user. Thus the app is quite like many AI prompting code editors, like Cursor, Zed, and Antigravity. The UX of playing with the app is however far different than other media-generating AI, for the user has total control of the final product via code editing, and the AI can easily iterate on previous structured output. The site design is a react-like SPA. The current designs are primarily to engage in onboarding of new users. The prompting of AI to generate code reduces the learning curve. Meanwhile, I have also designed the “system prompt”, or Agent, which is key to the quality of output, allowing users to simply prompt for a song with as much or little context as they may provide.

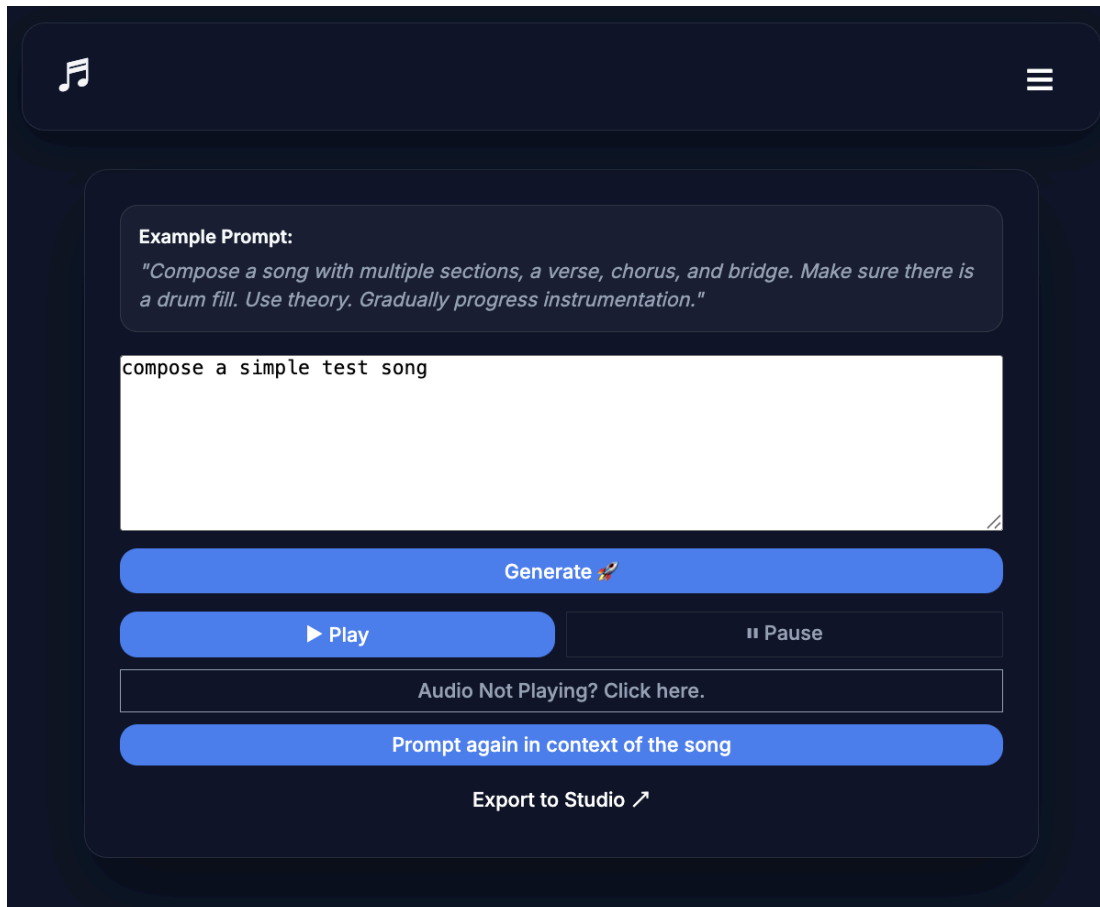
## A. Front Page - simple, demonstrative, and ready to access



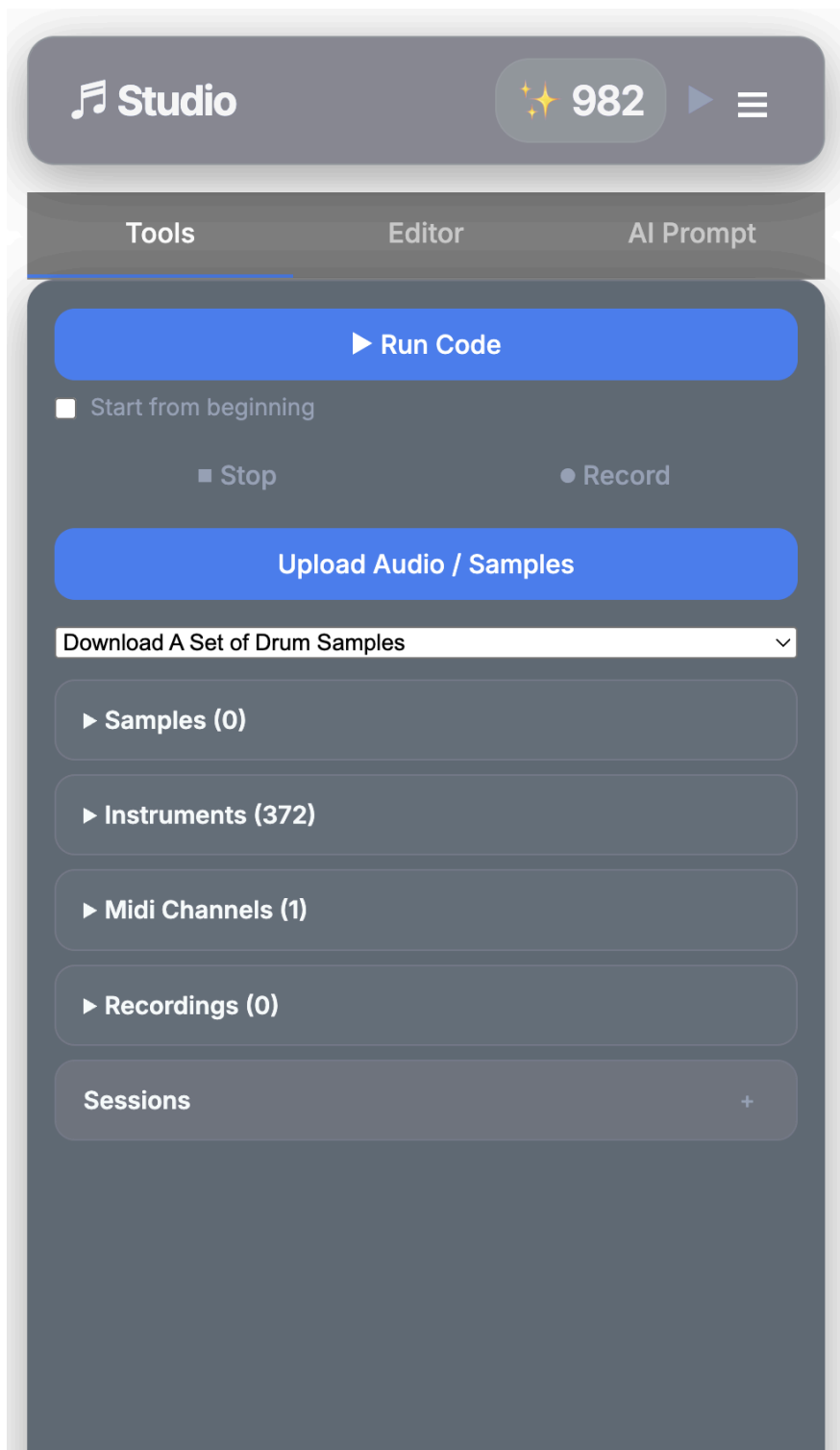
## B. Dashboard with payments



## C. A quick-prompt interface

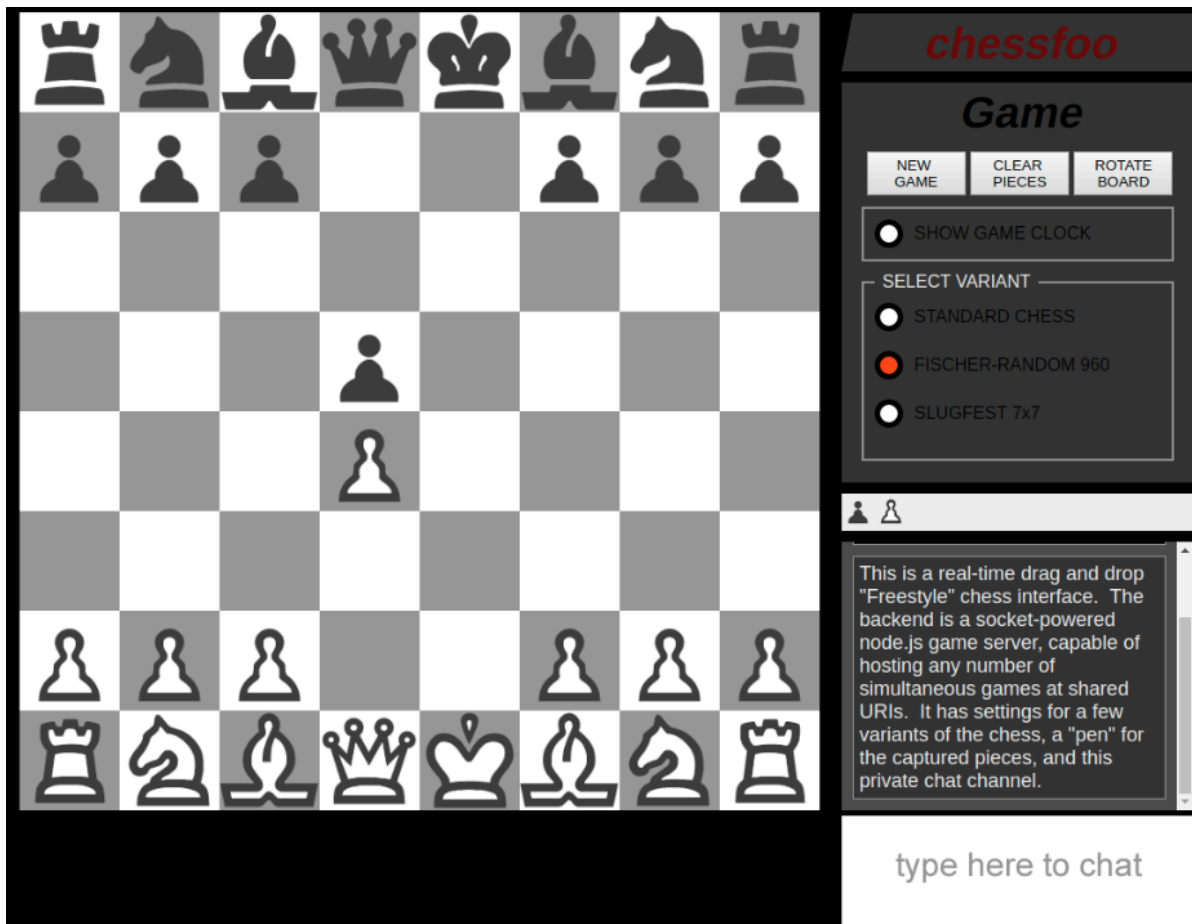


## D. The mobile studio UI



## 2. Chess Foo

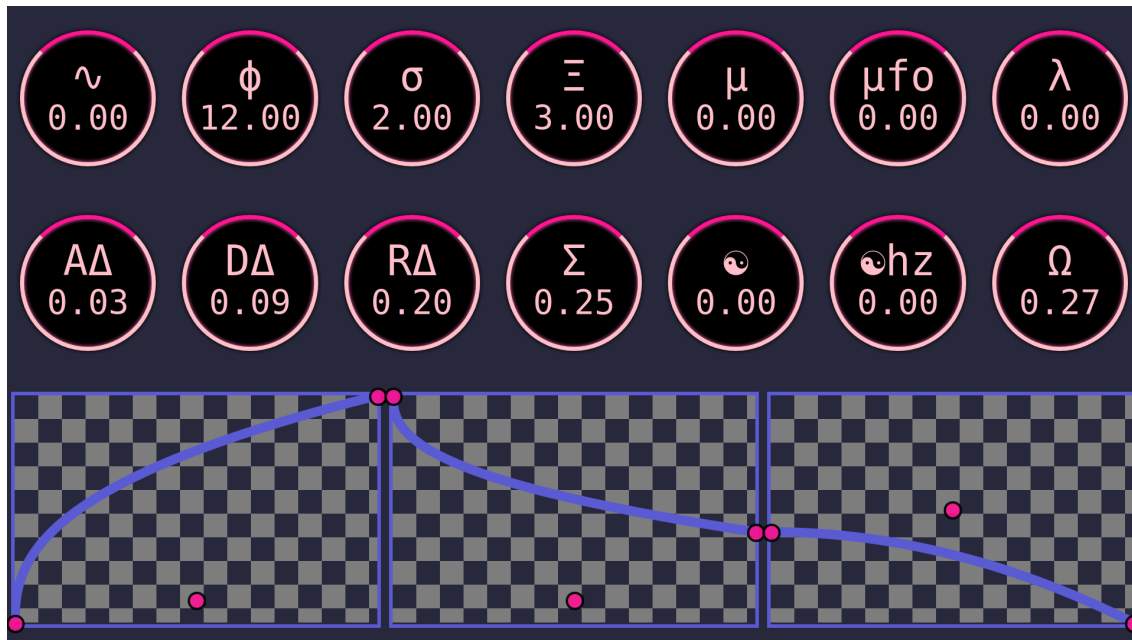
My first full stack app, built in 2013. This app comprises many design choices for the UI and UX. A large, clean game board and player pieces. Sidebar for options and chat. For UX I chose to allow pieces to be freely movable, not restricted to “legal moves” during a game, allowing players to set up their own challenges, or to analyze their game. The game also includes two variations of chess, one is called 960 Random. The tech stack for this game is websockets with a NodeJS backend, hosting games at unique URLs. Players are connected by a proxy for direct real-time play, chat, and game options.



### 3. Autogenerated Synthesizer Interface

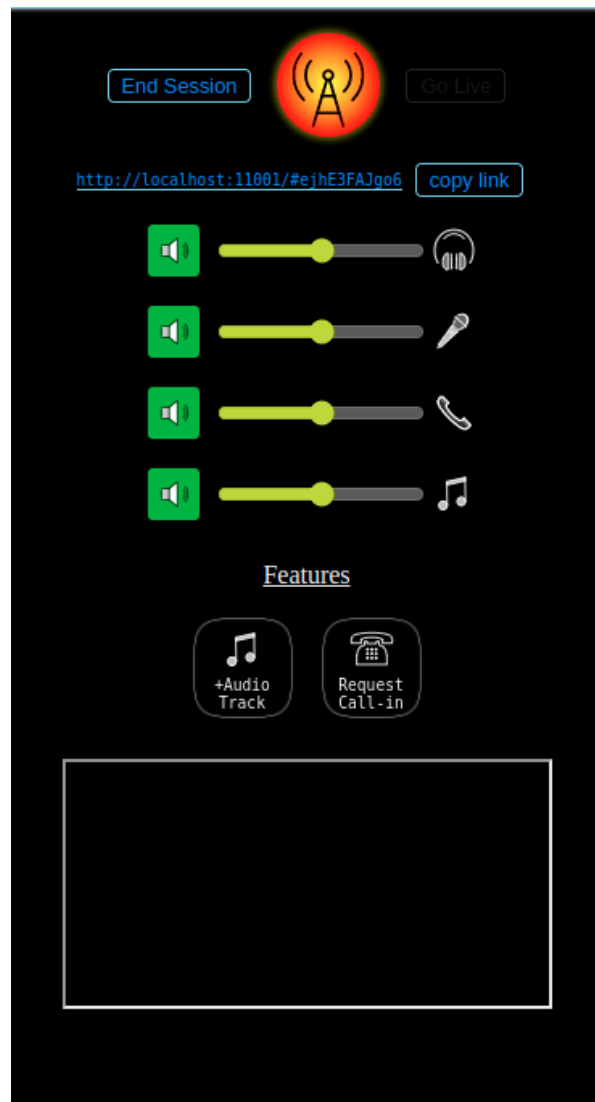
This interface and UX design is innovative and unique. All of the components are autogenerated from a configuration. They are designed to be precise and perform perfectly in a “live” system, in this case the continuous flow of audio generated by playing. This design transcends the app itself, which is why the UI is generated by a reusable module. React was still new when I built this. The UI and its components are designed for interacting with processes scientific, computational, and mathematical. To give the user real-time feedback when changing parameters. The round knobs emulate rotary dials found on hardware, which spin without end in either direction. The graphs are bezier curves defined by enveloping functions for the audio,

The science and tech of the system is worthy of note for its algorithms, mathematics, multimodal synthesis, and web API usage (webMidi, webAudio). The UI module is also highly technical.



#### 4. Mobile App for P2P Audio Broadcasting

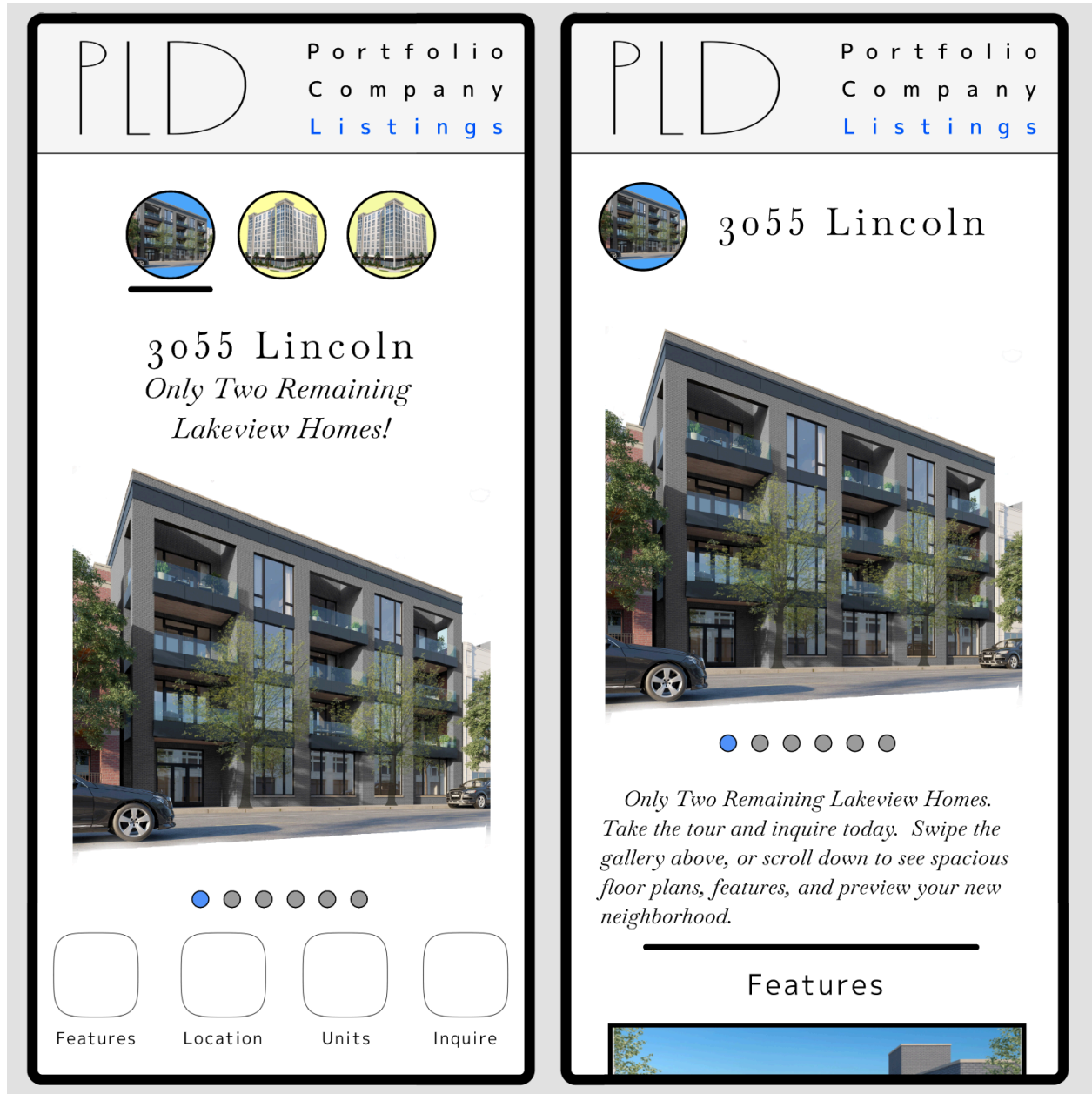
Another web app loaded with high performance tech, such as WASM Opus encoding, webRTC, webAudio, and all of it working in mobile Safari and old Android devices. The aim for this design was “international” which is why it uses “emoji” style glyphs instead of text. A simple mobile interface and app allows the host to broadcast voice, music and call-in audio from listeners. The backend used a signaling server for the webRTC.



## 5. Design With Figma

### A. Mobile Design - Home Construction Company Website

Intended to have the look and feel of a social feed site like instagram, for an intuitive user experience.





## B. Tablet Design for Information Rich Biology App

The challenge for this design was to fit a high density of data on the page along with graphical and interactive elements. I used reddit style comment collapsing to partition textual data. The bottom right module is a stand in for more interactive elements.

